

Toward Bi-directional In Situ Visualization and Analysis of Blood Flow Simulations With Dynamic Deforming Walls

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Abstract

We present some preliminary work toward bi-directional in situ visualization and analysis of blood flow considering blood cell suspensions and dynamic moving walls to mimic the clot growth process. The work is implemented through integrating multiple open-source packages including fluid flow simulation, blood cell modeling, and in situ visualization and analysis. It shows that in situ visualization enables us to examine and process the simulation results without writing data to files. In addition, preliminary results on bidirectional steering (still in progress) can accelerate engineers to explore the complicated stress environment experienced by clots with different sizes

Motivation

- Streamlining analysis for large scale simulations
- Enabling dynamic domain modification for in situ blood flow simulation
- Instrumenting an interactive in situ simulation platform for biomedical applications such as quick evaluation of embolism risk of a clot.

Results

- A nearly complete adaptation of prior singleCell showcase to work with SENSEI 4.0.0 for the purpose of enabling bidirectional component of in situ framework
- A workflow for creating a bidirectional in situ pipeline was identified. Figure 1 describes our bidirectional in situ pipeline and workflow
- Dynamic domain definition in Palabos extended to work with solid solver LAMMPS as seen in Figure 2

Future Work

- Expanding showcases to include large scale simulations
- GPU implementation
- Use of off-lattice boundary conditions
- Coupling with 1D solver for blood flow at arterial scale

Methods

We're using SENSEI [1] with Palabos/LAMMPS coupling code [2], building on top of the previous work [5]

LAMMPS

- Large Scale Atomic/Molecular Massively Parallel Simulator [4]
- Accepts user inputs such as bond type, number of particles, internal velocity, etc.
- Extended to model red blood cells with interactive membrane particles

Palabos

- Parallel Lattice Boltzmann Solver [3]
- Solves fluid (blood plasma) properties using Lattice Boltzmann method.
- Fluid velocity, vorticity, and normalized velocity

Bidirectional Instrumentation

- We updated the existing singleCell example to run with SENSEI 4.0 and ParaView 5.10
- The bidirectional preparation was done following the bidirectional oscillator example [6]
- The singleCell example has been modified to feature dynamic fluid boundary. The boundary is defined by a parametric function, values for which will later be modified from ParaView GUI
- The dynamic domain implementation was done using the data processor to first define the initial boundary conditions via bounceback nodes to then modify them in accordance with the parametric function mentioned above

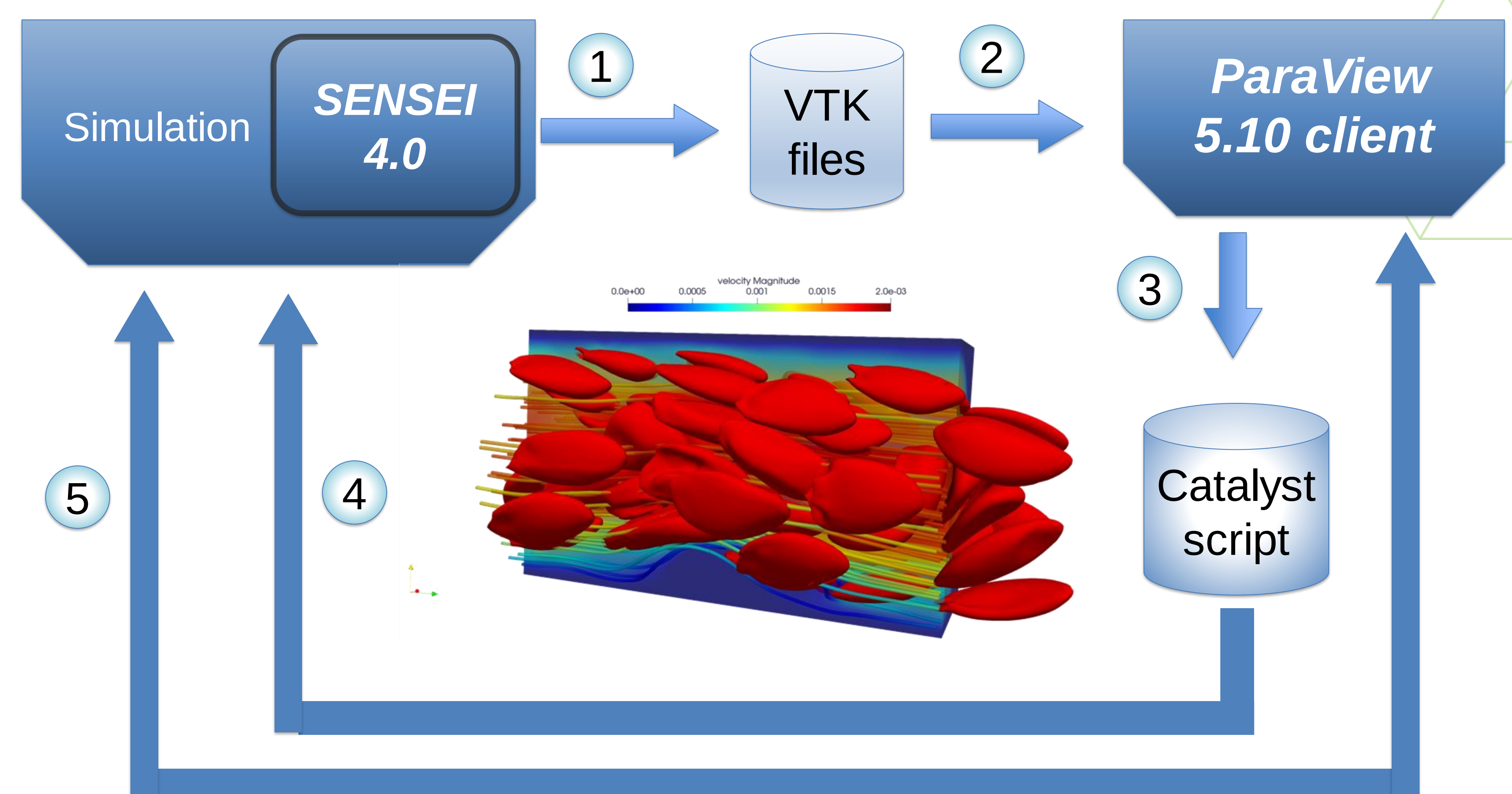


Figure 1: Workflow to create a bidirectional in situ visualization pipeline.

Workflow

1. Run Palabos/LAMMPS simulation for a few timesteps with SENSEI and save a representative dataset;
2. Visualize interactively with ParaView and define parameters for the visualization pipeline;
3. Save the pipeline as a Catalyst script after adding a PNG extractor;
4. Pass Catalyst script to simulation instrumented with SENSEI;
5. Visualize the simulation in situ; modify the simulation parameters in situ from the ParaView client.

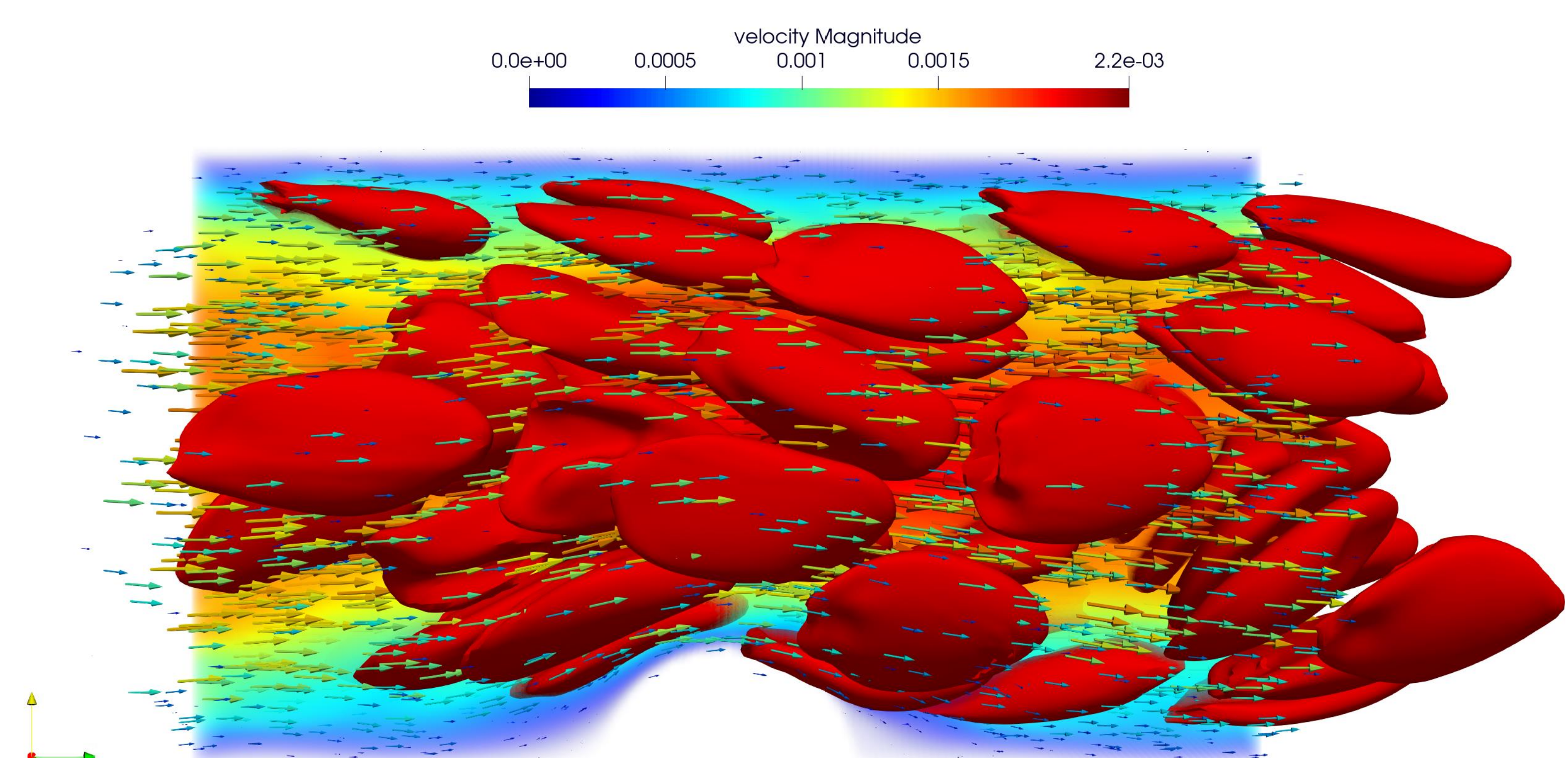


Figure 2: In situ visualization of blood flow in cell suspensions (red) with dynamically moving walls mimicking the clot growth process. The clot was modeled as a semi-cylinder at the center bottom of the vessel. Arrows are showing the velocity directions in the blood vessel.

This work was supported in part by the Director, Office of Science, Office of Advanced Scientific Computing Research, of the U.S. Department of Energy under Contract DE-AC02-06CH11357, through the grant "Scalable Analysis Methods and In Situ Infrastructure for Extreme Scale Knowledge Discovery". This work was supported by Northern Illinois University.

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